

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

1 a) Create a dept table with the following:

Did int(3) primary key

Dname varchar(10)

b) Insert two records in the dept table

c) Display the contents of the table

```
MariaDB [22co11]> create table dept
-> (Did int(3) primary key, Dname varchar(10));
Query OK, 0 rows affected (0.03 sec)

MariaDB [22co11]> describe dept;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Did   | int(3)        | NO   | PRI | NULL    |       |
| Dname | varchar(10)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

MariaDB [22co11]> insert into dept
-> values(10,'computer'),(11,'Mechanical');
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0

MariaDB [22co11]> select * from dept;
+-----+-----+
| Did | Dname   |
+-----+-----+
| 10  | computer |
| 11  | Mechanical |
+-----+-----+
2 rows in set (0.00 sec)
```

2 Create table emp1_rollno with the constraint applied at the column level.

Eid Int(3) Primary key

Ename Varchar(20) Not null

Doj Date Default system date

Eregno Int(5) Unique

Age Int(3) Age>=25

Dno Int(3) Foreign key

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```
MariaDB [22co11]> create table emp1_22co11
-> (Eid int(3) primary key, Ename varchar(20) not null, Projno int(3) default 2, Eregno int(5) unique, Age int(3) check(age>=25), Dno int(3) references dept(Did));
Query OK, 0 rows affected (0.03 sec)
```

3 Describe emp1_rollno table

```
MariaDB [22co11]> describe emp1_22co11;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Eid   | int(3)        | NO   | PRI | NULL    |       |
| Ename | varchar(20)   | NO   |     | NULL    |       |
| Projno | int(3)        | YES  |     | 2       |       |
| Eregno | int(5)        | YES  | UNI | NULL    |       |
| Age   | int(3)        | YES  |     | NULL    |       |
| Dno   | int(3)        | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.02 sec)
```

4 Create table emp2_rollno with the constraint applied at the table level.

Eid Int(3) Primary key

Ename Varchar(20) Not null

Doj Date Default system date

Eregno Int(5) Unique

Age Int(3) Age>=25

Did Int(3) Foreign key referring

dept(did)

```
MariaDB [22co11]> create table emp2_22co11
-> (Eid int(3), Ename varchar(20) not null, Projno int(3) default 2, Eregno int(5), Age int(3), Did int(3), primary key(Eid), check(age>=25), foreign key(Did) references dept(Did));
Query OK, 0 rows affected (0.03 sec)
```

5 Describe emp2_rollno table

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```
MariaDB [22co11]> describe emp2_22co11;
```

Field	Type	Null	Key	Default	Extra
Eid	int(3)	NO	PRI	NULL	
Ename	varchar(20)	NO		NULL	
Projno	int(3)	YES		2	
Eregno	int(5)	YES		NULL	
Age	int(3)	YES		NULL	
Did	int(3)	YES	MUL	NULL	

```
6 rows in set (0.02 sec)
```

6 Create table emp3_rollno with all the constraints that can be applied at the table level with constraint names

Eid Int(3) Primary key

Ename Varchar(20) Not null

Doj Date Default system date

Eregno Int(5) Unique

Age Int(3) Age>=25

Did Int(3) Foreign key referring

dept(did)

```
MariaDB [22co11]> create table emp3_22co11
-> (Eid int(3), Ename varchar(20) not null, Projno int(3) default 2, Eregno int(5), Age int(3), Did int(3), constraint pk_eid primary key(eid), constraint u_Eregno u
nique(Eregno), constraint c_age check(age>=25), constraint fk_did foreign key(Did) references dept(Did));
Query OK, 0 rows affected (0.03 sec)
```

7 Describe emp3_rollno table

```
MariaDB [22co11]> describe emp3_22co11;
```

Field	Type	Null	Key	Default	Extra
Eid	int(3)	NO	PRI	NULL	
Ename	varchar(20)	NO		NULL	
Projno	int(3)	YES		2	
Eregno	int(5)	YES	UNI	NULL	
Age	int(3)	YES		NULL	
Did	int(3)	YES	MUL	NULL	

```
6 rows in set (0.02 sec)
```

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8 Create table emp4_rollno with fields given below without any constraints

Eid Int(3)

Ename Varchar(20)

Doj Date

Eregno Int(5)

Age Int(3)

Did Int(3)

```
MariaDB [22co11]> create table emp4_22co11
-> (Eid int(3), Ename varchar(20), Projno int(3), Eregno int(5), Age int(3), Did int(3));
Query OK, 0 rows affected (0.01 sec)
```

9 Define the constraints using alter table commands for the emp4_rollno table

a) Add a primary key

```
MariaDB [22co11]> alter table emp4_22co11
-> add primary key(Eid);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

b) Modify ename to not null and size 25

```
MariaDB [22co11]> alter table emp4_22co11
-> modify Ename varchar(25) not null;
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

c) Add unique constraint along with constraint name to eregno

```
MariaDB [22co11]> alter table emp4_22co11
-> add constraint uniqueregno unique(Eregno);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

d) Drop unique constraint from eregno

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```
MariaDB [22co11]> alter table emp4_22co11
-> drop index uniqueregno;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

10 Insert 7 records to verify the constraints

a) Primary key violation

```
MariaDB [22co11]> insert into emp3_22co11 values
-> (1,'Jack',20, 500, 34, 11);
Query OK, 1 row affected (0.00 sec)

MariaDB [22co11]> insert into emp3_22co11 values
-> (1,'Joan',21, 502, 32, 10);
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'
```

b) Unique key violation

```
MariaDB [22co11]> insert into emp3_22co11 values
-> (2,'Jaden',22, 502, 56, 10);
Query OK, 1 row affected (0.00 sec)

MariaDB [22co11]> insert into emp3_22co11 values
-> (3,'Jerald',22, 502, 50, 10);
ERROR 1062 (23000): Duplicate entry '502' for key 'u_Eregno'
```

c) Check constraint violation

```
MariaDB [22co11]> INSERT INTO emp3_22co11 (Eid, Ename, Eregno, Age, Did) VALUES (4, 'Eve', 12347, 22, 101);
ERROR 4025 (23000): CONSTRAINT `c_age` failed for `22co11`.`emp3_22co11`
```

d) Not null constraint violation

```
MariaDB [22co11]> INSERT INTO emp3_22co11 (Eid, Ename, Eregno, Age, Did) VALUES (5, NULL, 12348, 35, 102);
ERROR 1048 (23000): Column 'Ename' cannot be null
```

e) Default constraint

```
MariaDB [22co11]> INSERT INTO emp3_22co11 (Eid, Ename, Eregno, Age, Did) VALUES (6, 'Frank', 12349, 36, 10);
Query OK, 1 row affected (0.071 sec)
```

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```
MariaDB [22co11]> select * from emp3_22co11;
```

Eid	Ename	Projno	Eregno	Age	Did
1	Jack	20	500	34	11
2	Jaden	22	502	56	10
6	Frank	2	12349	36	10

3 rows in set (0.002 sec)

f) Foreign key constraint

```
MariaDB [22co11]> INSERT INTO emp3_22co11 (Eid, Ename, Eregno, Age, Did) VALUES (7, 'Grace', 12350, 27, 999);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`22co11`.`emp3_22co11`, CONSTRAINT `fk_did` FOREIGN KEY (`Did`) REFERENCES `dept` (`Did`))
```

g) Correct record entry

```
MariaDB [22co11]> INSERT INTO emp3_22co11 (Eid, Ename, Eregno, Age, Did) VALUES (7, 'Grace', 12350, 27, 10);
Query OK, 1 row affected (0.212 sec)
```

```
MariaDB [22co11]> select * from emp3_22co11;
```

Eid	Ename	Projno	Eregno	Age	Did
1	Jack	20	500	34	11
2	Jaden	22	502	56	10
6	Frank	2	12349	36	10
7	Grace	2	12350	27	10

4 rows in set (0.003 sec)